AHMAD DAFA SALAM

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PROFESSIONAL SUMMARY

Computer Engineering undergraduate at Institut Teknologi Sepuluh Nopember (ITS) with production software engineering experience across full-stack web platforms (FastAPI, Node.js, React, PostgreSQL), cross-platform mobile applications (Flutter), computer vision and OCR automation (OpenCV, RapidOCR, deterministic DOS character-grid matching, ONNX Runtime), and embedded IoT systems (ESP32, C++). Proven track record of diagnosing and resolving mission-critical accuracy and concurrency bottlenecks, automating enterprise HR and certification workflows at PT PLN Nusantara Power Services, and delivering end-to-end tested systems from hardware firmware to cloud-deployed architectures.

EDUCATION

Institut Teknologi Sepuluh Nopember (ITS) Surabaya, Indonesia
Bachelor of Engineering in Computer Engineering (FT-EIC) Aug 2023 – Present
GPA: 3.50/4.00 (Latest semester GPA: 3.93/4.00)

PROFESSIONAL EXPERIENCE

PT PLN Nusantara Power Services Surabaya / Sidoarjo, Indonesia
Software Engineering & IT Intern – Human Capital & Training Development Center (TDC) Oct 2026 – Present
Python, FastAPI, PostgreSQL (Neon), SQLAlchemy, Alembic, RapidOCR, PyMuPDF, OpenPyXL, Vercel

- Architected and deployed an enterprise Certification & Training Monitoring Platform tracking 1,137 standardized competency codes (`Resource Code` & `Level 1-9 SKP`), separating 3/5-year expiring technical certifications from non-expiring training records across generation units.
- Engineered an idempotent two-stage expiration notification service (`H-30` and `H-20` automated email alerts with CC to TDC Planning), a 4-format header-detecting corporate Excel importer, and monthly Ellipse HR mutation synchronization to reconcile employee transfers and promotions.
- Built **Certificate Manager v3.0** (`cert-slicer`), an offline-first PDF certificate slicer and 3-database employee matcher combining PyMuPDF vector parsing, multi-threaded RapidOCR, and Gemini 2.5 Flash fallback with an 82,989-file Synology NAS/FTP archive index.

LAB M-IOT Surabaya, Indonesia
Software Engineering Intern – MMPI-2 Automated Psychological Assessment Platform Sep 2026 – Present
Python, OpenCV, Deterministic 80×25 DOS Grid Vision, Win32 API, PostgreSQL, Docker

- Diagnosed and resolved a critical 74.0% (`185/250` scales) scoring accuracy bottleneck in the 566-item MMPI-2 clinical assessment platform caused by legacy Tesseract OCR glyph confusion (`8` vs `0`, `6` vs `5`) and narrow column slicing that truncated extreme clinical T-scores (`> 100` and `> 120`).
- Engineered a deterministic 80×25 DOS character-cell bitmap matcher (`baca_layar.py`) with `SetProcessDPIAware()` calibration and widened column extraction, achieving **100.0% scoring accuracy (`250/250` clinical and research scales across 10 synthetic test patterns)** with zero external OCR dependency.
- Implemented pre-scoring 566-item `+/-` screen buffer verification prior to `Esc` commit, PID-bound Win32 window focus locking to prevent unfocused `pyautogui` keystroke leaks, and `host.docker.internal` container bridge stabilization.

PROJECTS

SmartCoop – Poultry Health & Biosecurity Management System 2026
Flutter, Supabase (PostgreSQL/RLS), FastAPI, ONNX Runtime, Docker, Railway, NFC

- Delivered a cross-platform mobile application (Flutter + Supabase) that detects poultry diseases from droppings or physical condition photos through a three-model ONNX pipeline served by a containerized FastAPI backend.
- Designed a 12-table PostgreSQL schema on Supabase with Row Level Security (RLS) for strict multi-farm data isolation and implemented an 8-phase guided biosecurity SOP with per-bird medical records linked via unique NFC tags.

Planner System 4.0 – Contract-First Multi-Agent Swarm & Verification Engine 2026
Python, SQLite WAL Blackboard, AST Mutation Testing, HMAC-SHA256 Sealed Oracles, APCA Contrast Kinematics

- Built a deterministic multi-agent software engineering orchestrator coordinating 8 parallel worker accounts with strict port isolation (`3000-3799` / `8000-8799`), atomic SQLite WAL ticket leasing, and automatic Chameleon quota failover.
- Enforced Bicameral Zero-Trust verification: cryptographically signed HMAC-SHA256 Sealed Oracles isolated from builder agents and an AST Saboteur gate that injects synthetic code mutations to enforce a $\geq 80\%$ mutation kill score across 7/7 Golden Benchmarks.

Hermes JobSeeker Bot v2.0 – ITS Career Center Scraper, Honest Auto-Apply & Smart Inbox Tracker 2026
Python, Asyncio, Playwright CDP, Gemini Vision OCR, myITS Entra ID 2FA Bridge, Telegram API, 3NF SQLite WAL, Pytest (85/85 PASS)

– Engineered an autonomous job intelligence engine scraping `careers.its.ac.id`, `@itscareercenter` Instagram flyers (via a 4-layer anti-block pipeline, dHash deduplication, and Gemini Vision OCR), LinkedIn, JobStreet, Glints, and Kalibrr into a normalized 3NF SQLite WAL database (196 verified jobs).
– Built a Multi-Channel Honest Auto-Apply Engine supporting direct SMTP email, `careers.its.ac.id` myITS Microsoft Entra ID SSO with live 2-digit Authenticator Number Matching (`#idRichContext_DisplaySign`) bridged to Telegram, and a Smart HR Inbox Manager classifying replies into Offered, Interview, Assessment, and Rejected.

Warung Madura Enterprise POS & Multi-Unit Inventory Manager (DevSprint 3.0) 2026
Node.js, Express, TypeScript, React 18, Tailwind CSS, Role-Based Access Control (RBAC)

– Developed a 24-hour FMCG retail ERP and keyboard-first Express POS (`Sat-Set`) supporting hierarchical packaging conversions (`Eceran`, `Pak`, `Slop`, `Dus`, `Karung`), atomic stock deduction, and 58mm thermal receipt printing.
– Enforced three-tier RBAC (`Owner`, `Warehouse`, `Cashier`) with API-level wholesale cost (`harga_beli`) masking, smart safety-stock reorder proposals, and full Purchase Order lifecycle management.

Freedom v2.0 (Speed Edition) – USB Encrypted Tunneling & Hardware-Locked Windows Daemon 2026
PowerShell, Windows Service Manager, Sing-box / Xray-Core, SMBIOS WMI Hardware Cryptography, Batch CLI

– Created a 1-click USB-deployable network provisioning suite installing a self-healing Windows service (`WinNetOptimizerSvc`) that routes traffic through a multiplexed TLS 1.3 / SOCKS5 edge tunnel on restricted networks.
– Implemented deterministic anti-cloning hardware binding via `SHA-256(Motherboard UUID | Hostname | Salt)` and stripped 35 MB GeoIP/GeoSite startup overhead to cut memory usage by 68% with sub-35ms handshake latency.

Big Data Intelligence Analysis Pipeline & RAG Telegram Chatbot 2025
Python, n8n, Apify, Groq API (Llama 3.3 70B), Supabase, Chart.js

– Built an automated end-to-end pipeline (web crawling → LLM sentiment, emotion, and key-point extraction → PostgreSQL storage) paired with a Telegram RAG chatbot and real-time Chart.js analytics dashboard.

Card Detective – Real-Time Card Recognition Game 2025
Python, OpenCV, NumPy

– Implemented a real-time computer vision game using pure OpenCV (Canny edge detection, contour filtering, 4-point perspective warp, and pixel template matching) with a complete state machine and live HUD preview.

Autonomous Driving Car & SmartPot Predictive IoT Watering System 2025
ESP32, C++, HC-SR04 Ultrasonic Array, L298N H-Bridge, Soil/DHT/LDR Sensors, Blynk IoT

– Built an ESP32 autonomous robot with 270° 5-sensor ultrasonic spatial awareness and differential-drive steering, alongside an IoT plant watering system featuring a 1-hour-ahead evaporation soil-dryness prediction model.

SKILLS

– **Programming Languages:** Python, TypeScript, JavaScript (ES6+), C/C++, Dart, SQL, PowerShell, Bash
– **Backend & Web Development:** FastAPI, Node.js, Express.js, React 18, Flutter, RESTful APIs, HTML5/CSS3, Tailwind CSS
– **AI, Computer Vision & Automation:** OpenCV, RapidOCR, ONNX Runtime, Playwright, LLM Orchestration (Gemini / Llama 3.3), n8n
– **Databases & Cloud Infrastructure:** PostgreSQL (Neon / Supabase), Row Level Security (RLS), SQLite (WAL), SQLAlchemy, Alembic, Docker, Vercel, Railway, Git/GitHub
– **Embedded Systems & Hardware:** ESP32, Arduino, Ultrasonic/Environmental Sensor Fusion, NFC (ISO14443A), L298N Motor Control
– **Languages:** Indonesian (Native), English (Professional Working Proficiency)

ORGANIZATIONAL & COMMITTEE EXPERIENCE

Multimedia and Game Event (MAGE) XI Surabaya, Indonesia
Expert Staff (Staf Ahli), Event Division Mar 2025 – Nov 2025

– Contributed to MAGE XI, a national-scale IT competition organized by the Computer Engineering Department at ITS covering Multimedia, Robotics, and IoT categories, coordinating budget planning (RAB), event rundowns, and staff supervision.

HIMATEKKOM ITS (Himpunan Mahasiswa Teknik Komputer) Surabaya, Indonesia
Staff of Student Welfare Division (Kesejahteraan Mahasiswa / KESMA) 2025

– Acted as student welfare liaison between Computer Engineering students and the department, and initiated JAVA (*Jaringan Advokasi*), an online advocacy and academic feedback channel.